

Queue Theory: Part II

ORSA Study text



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Queuing Models – Waiting lines

MEASURES OF PERFORMANCE

There are various ways of judging how well a queuing system is performing. Results may be evaluated over a short period of time once the system opens, or based on the long run or equilibrium results.

- Time customers spend waiting (average waiting time or at a measure such as the percent of customers who have to wait longer than t minutes).
- The average number of customers in line.
- The utilisation rate of servers.
- Cost of operating the system.

The manager would use these measures to determine the optimum number of servers and capacity for the queuing system.

A given queuing system can have any combination of the elements described above. Hence, there are a very large number of possible systems, and no one mathematical model can describe them all. In this section, we focus on a few simple models that have wide applicability, and that give us insight into queuing system behaviour in general. Most queuing systems behave like the cases examined in this chapter.

There are various ways of judging how well a processing system is performing. Results may be evaluated over a short period of time once the system opens, or they may be based on the long-run or equilibrium results. Generally, the time jobs spend waiting is important, and we may look at the average waiting time or at a measure such as the percent of jobs that wait longer than, say, 10 minutes. A related measure is the *throughput time* for a job (waiting time plus service time). The length of the waiting line is another common measure of performance. These are measures of how well the system is performing from the customer point of view.

Other measures relate to the cost of operating the system. The system load factor or capacity utilisation measures the ability of the system to handle the arrival load. Management has the option of adding more capacity.

MODELLING OF RANDOM RELATIONS IN THE QUEUE

There are two important probability distributions that appear often in queuing models. The Poisson distribution assumes a very large (infinite) number of possible arrivals, with each having a small probability of occurrence. Arrivals are independent and the number of arrivals in one period of time does not affect the number in the next. The Poisson distribution is a good approximation for many processes that are considered "random," such as calls arriving at a switchboard, customers at a check-in counter, or fire alarms.

The random variable in the Poisson distribution is the *number of events (e.g., arrivals) in a unit of time*. The exponential distribution is a complementary distribution to the Poisson that has as the random variable the time between events. Both of these distributions are called Markov distributions.

A SINGLE-SERVER QUEUING MODEL

Kendall's Notation for Queues

Kendall's notation for queues is a useful way to represent different types of queues in a compact and easily understood fashion. Kendall's notation describes:

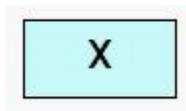
1. nature of the arrival process to the queue
2. nature of the service time
3. number of servers
4. maximum number customers in the queue
5. queuing disciplines (behaviour of customers in the queue)

The notation has been considerably extended to allow it to represent a wide variety of queues.

Following this representation, a queue is represented by a sequence

X/Y/k/Q/D

with the following meaning attached to the letters A to E:



"X" symbolically represents the nature of the arrival process to the queue. Special letters are used to symbolise the nature of the interarrival time distribution as follows:

- M Exponentially distributed interarrival times (Poisson Process, Markov Process)
- D Deterministic (fixed) inter-arrival times
- Ek Erlang distribution of order k for the inter-arrival times
- Hk Hyper-exponential distribution of order k for the inter-arrival times
- G General (any!) distribution for the inter-arrival times
- etc.



"Y" symbolically represents the nature of the service time distribution for the customers getting served in the queue. The same letters as the ones above are used to describe the nature of the service time distribution.



"k" Number of servers in the queue



"Q" Maximum number of customers that can be there in the queue - this includes both the ones currently being served and the ones waiting for service. Default is infinity many which is assumed when this is omitted.



"D" Queuing discipline:

FCFS	First	Come	First	Served
LCFS	Last	Come	First	Served
SIRO	Service	in	random	order
etc.				

Example

M/M/1

Poisson Arrivals, Exponential Service Time Distribution, Single Server, Infinite Number of Waiting Positions

M/Ez/2/K

Poisson Arrivals, Erlangian of order-2 Service Time Distribution, 2 Servers, Maximum Number K in queue

G/M/2

Generalised Arrivals, Exponential Service Time Distribution, 2 Servers, Infinite Number of Waiting Positions

MATHEMATICAL MODEL OF M/M/1 SYSTEM

Model Description

1. Arrivals are random, and come from the Poisson probability distribution (Markov impute).
2. Each service time is also assumed to be a random variable following the exponential distribution (Markov service).
3. Service times are assumed to be independent of each other and independent of the arrival process.
4. There is one single server in the queue.
5. The queue discipline is FIFO, and there is no limit on the size of the line.
6. The average arrival and service rates do not change over time. The process has been operating long enough to remove effects of the initial conditions.

Notation

We denote:

λ	Average number of customers arriving in one unit of time
μ	Average number of customers the facility is capable of servicing in one unit of time, assuming no shortage of customers
L	Expected number of units being serviced and/or waiting in the system
L_q	Expected number in the queue (the number in the queue does not include the unit being serviced)
p_n	Probability of having n units in the system
W_q	Expected time an arrival must wait in the queue
W	Expected time an arrival spends in the system (both in queue and in service)

Basic numeric characteristics of M/M/1 Queue

System must be operating long enough so that the probabilities resulting from the physical characteristics of the problem may satisfy the requirements of the mass selection of a statistical observation - that is, the system must be in equilibrium.

Define $\rho = \frac{\lambda}{\mu}$ as an utilisation of the system and suppose $\rho < 1$. There are following relationships of interest:

1. Probability that there are n arrivals in the system (in case of system M/M/1 there is one arrival in the service channel and the rest $n-1$ arrivals are waiting in the queue):

$$p_n = \rho^n(1 - \rho)$$

2. Probability of the system being empty - expected idle time of the system:

$$p_0 = (1 - \rho)$$

3. Probability of the system being not empty - expected busy time of the system or expected utilisation:

$$1 - p_0 = \rho$$

4. The expected number in the waiting line and/or being serviced is:

$$L = \frac{\lambda}{\mu - \lambda}$$

5. The expected number in the queue is:

$$L_q = \frac{\lambda^2}{\mu(\mu - \lambda)}$$

6. The average waiting time (in the queue) of an arrival is:

$$W_q = \frac{L_q}{\lambda} = \frac{L}{\mu}$$

7. The average time an arrival spends in the system (both waiting and in service) is:

$$W = W_q + 1/\mu$$

The probability that the number in the queue and being serviced is greater than M is:

$$P\{n > M\} = \left(\frac{\lambda}{\mu} \right)^{M+1}$$

Example

Analyse system of queue in an office. Population of customers is very large so that we can consider it is an infinite one). Customers for the service arrive randomly following a Poisson process. The office can process customers at an average rate of five patients an hour (one at a 12 minutes). The service process is also Poisson. Customers are served at an average of four per hour (one at a 15 minutes). The office operates 12 hours a day.

First, all measures assume the process has been operating long enough for the probabilities resulting from the physical characteristics of the problem to have made themselves felt; that is, the system is in equilibrium.

Second, the utilisation of the system is $r = \lambda/\mu = 4/5 = 0,8 < 1$, thus we can count relationships as follows:

1) Probability of the system being empty - expected idle time of the system:

$$p_0 = (1 - r) = 1 - 0,8 = 0,2$$

On average, the office will be idle 20 percent of the time and busy 80 percent of the time.

2) The expected number in the system - both in waiting line and being serviced is:

$$L = \frac{\lambda}{\mu - \lambda} = \frac{4}{5 - 4} = 4$$

There will be an average of four persons in line and being serviced.

3) The expected number in the waiting line is:

$$L_q = \frac{\lambda^2}{\mu(\mu - \lambda)} = \frac{4 \cdot 4}{5(5 - 4)} = \frac{16}{5} = 3,2$$

There will be an average of 3,2 people in waiting line.

4) The average waiting time (in the queue) of an arrival is:

$$W_q = \frac{L_q}{\lambda} = \frac{L}{\mu} = \frac{16/5}{4} = \frac{4}{5} = 0,8$$

The average waiting time of an customer is 0,8 of an hour (e.g. 48 minutes).

5) The average time an arrival spends in the system (both waiting and in service) is:

$$W = W_q + 1/\mu = 0,8 + 1/5 = 1$$

The average time spent in the system - both in the waiting line and obtaining service - is one hour.

If we assume a 12 hour workday, there will be an average of 48 customers arriving per day, and the expected total lost time of customers waiting will be:

$$T = \lambda \cdot 12 \cdot W_q = 4 \cdot 12 \cdot 0,8 = 38,4.$$

There will be a cost associated with these 38,4 hours. Assume the cost to the society is \$10 for each hour lost by a customer waiting. The average cost per day from waiting is

$$38,4 \cdot 10 = \$384$$

Suppose that we could in some fashion increase the service rate from five to six per hour and thereby decrease the average time spent in service from 12 minutes to 10 minutes. What would be the effect of this change?

With $\mu = 6$, the expected number in the queue is

$$L_q = \frac{\lambda^2}{\mu(\mu - \lambda)} = \frac{4 \cdot 4}{6(6 - 4)} = \frac{16}{12} = 1,33$$

instead of the 3.2 found above. The average wait for the patient is now:

$$W_q = \frac{L_q}{\lambda} = \frac{1,33}{4} = \frac{1}{3} = 0,33 \quad (\text{of an hour}).$$

Before the change, each customer spent an average of 12 minutes being served and 48 minutes waiting. After the change, each patient will spend an average of 10 minutes being served, and 20 minutes waiting. This can be verified by computing the total time in the system:

$$W = W_q + 1/\mu = 1/3 + 1/6 = 1/2 \quad (\text{of an hour}).$$

Each day there are 48 customers (4.12), and each customer will save one-half hour in total. At a cost of \$10 per hour, the daily cost savings is:

$$48 \cdot 1/2 \cdot 20 = \$480.$$

It would be worth \$480 per day or \$175200 per year to the society to increase the service rate to six customers at office per hour.

Queuing System Behaviour

The formulas presented for the single system M/M/1 case enable us to study system behaviour as the arrival rate λ approaches the service rate μ . Recalling that

$$\frac{\lambda}{\mu} = \rho$$

equals system utilisation or the percentage of time the server is busy, we can plot the expected number in the system (L) as a function of utilisation, see in Figure 1. By dividing the numerator and denominator of equation $L = \lambda / (\mu - r)$ by μ , we obtain number L as a function of utilisation ρ :

$$L = \frac{\frac{\lambda}{\mu}}{1 - \frac{\lambda}{\mu}} = \frac{\rho}{1 - \rho}$$

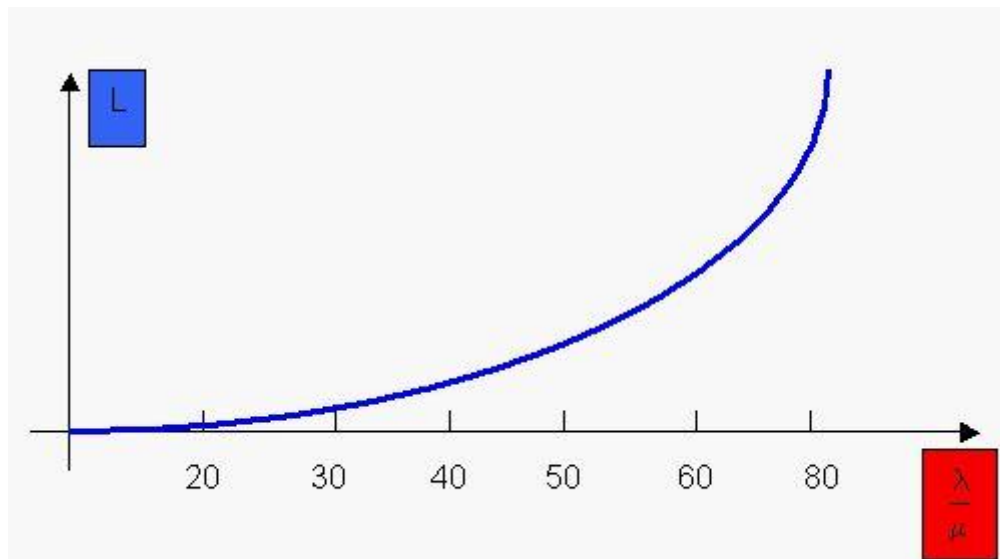


Figure 1

Figure 1 shows how responsive the number in the system L is to the utilisation r . In a single queuing system M/M1 as the utilisation r rises much above 80 percent, the expected number L in the system rises *ever more sharply*. The general implication is that if prompt service is desired in a single channel system, there must be parameter λ much less than parameter μ in order to avoid excessive waiting.

Although Figure 1 is directly applicable only under the stated assumptions, it is striking how many business situations behave in a qualitatively similar manner when utilisation changes. Even in complex production facilities with many machines and alternative routings for parts, it has often been observed that when increased demand leads to an increase in shop utilisation, there is an even greater increase in the time an order spends in the shop (waiting time plus time being processed), fitting the general pattern of Figure 1.

There is pressure in most organisations to be efficient and to reduce idle time as much as possible. Managers do not like to see personnel or equipment standing idle. Yet with random arrivals and service times, we see the need to have excess capacity to avoid long lines and long waits for customers. This is a very important insight about queuing systems and one that is not obvious to most people.

How to Analyse Random Data

Suppose you are operating a system in which customers arrive in random (Markov) process. Table 1 shows a history of 30 arrivals over a period of about eight hours. The left part of the table shows the actual times of arrival, when the customers actually started service, and when they were finished. The numbers at the right are calculated from these data: the length of waiting line ahead of the arrival, the interarrival time, the actual time taken to service the patient, and the amount of time each patient spent waiting.

Note that there is considerably variability in arrival patterns and in service time needed.

customer N ^o	History of the process				Proceeded data		
	arrival time	start service time	finish service time	length of queue	time since prior arrival	service time	waiting time
	hours				minutes		
0	8,00						
1	8,07	8,07	8,13	0	7	6	0
2	8,11	8,13	8,28	0	4	15	2
3	8,19	8,28	8,40	0	8	12	9
4	8,46	8,46	8,58	0	27	12	0
5	9,06	9,06	9,22	0	20	16	0
6	9,49	9,49	9,56	0	43	7	0
7	9,51	9,56	10,11	0	2	15	5
8	10,06	10,11	10,21	0	15	10	5
9	11,24	11,24	11,30	0	78	6	0
10	11,31	11,31	11,47	0	7	16	0
11	11,35	11,47	12,04	0	4	17	12
12	11,41	12,04	12,19	1	6	15	23
13	11,52	12,19	12,35	1	11	16	27
14	12,02	12,35	12,48	2	10	13	33
15	12,12	12,48	13,07	2	10	19	36
16	12,39	13,07	13,24	1	27	17	28
17	12,40	13,24	13,40	2	1	16	44
18	12,45	13,40	13,49	3	5	9	55
19	12,59	13,49	13,59	3	14	10	50
20	13,02	13,59	14,06	4	3	7	57
21	13,29	14,06	14,16	3	27	10	37
22	13,42	14,16	14,23	3	13	7	34
23	13,46	14,23	14,34	4	4	11	37
24	13,48	14,34	14,46	5	2	12	46
25	13,52	14,46	14,58	5	4	12	54
26	13,54	14,58	15,10	6	2	12	64
27	14,15	15,10	15,18	5	21	8	55
28	14,18	15,18	15,30	5	3	12	60

29	14,23	15,30	15,39	5	5	9	67
30	15,27	15,39	15,54	1	64	15	12
Average				2,00	14,90	12,10	28,40

Table 1: A history of 30 arrivals over a period of about eight hours

For example, seven customers arrived in the hour between 13:00 to 14:00 (1 to 2 p.m.) and yet there was only one arrival in the hour between 10:00 and 11:00. Similarly, some customers needed only 6 or 7 minutes of service time, while one patient took 19 minutes.

As can be seen, the first arrivals had zero or short waits. However, after customer number 11, the line and waiting times began to build up. Several arrivals had to wait more than 50 minutes, and the waiting line built up to as many as 6 customers by the time customer 26 arrived. But the line shortened again, so that customer number 30 had a short wait.

In the table one can see patterns that are typical in most queuing systems. There is variability in the interarrival times and in the service times for customers, and this results in lines and waits for service. Note that while the average waiting time was 28,4 minutes, this number doesn't fully capture the waiting time pattern. Eight of the 30 arrivals had to wait 50 minutes or more, and three people an hour or more.

In terms of service capacity, note that the table covered a period of 474 minutes. The average service time was 12,1 minutes. Thus, the total service time for all 30 arrivals was $30 \times 12,1 = 363$ minutes. Capacity utilisation was thus $363/474 = 77$ percent. The system was servicing customers only about three-fourths of the time. And yet there was substantial congestion with long waits for customers.

All data can be processed by suitable statistical software to obtain information about distributions of arrivals and service time (for example " χ -quadrate can be used) and corresponding parameters λ and μ .

Example of a M/M/1 Queue

A bank wants to know how many customers are waiting for a drive-in teller, how long they have to wait, the utilisation of the teller, and what the service rate would have to be so that 95 percent of the time there will not be more than three cars in the system at any one time.

Properties of the system:

Single server, infinite population, Poisson arrivals, Exponential service pattern, unlimited queue.

Typical situations: Drive-in teller at banks, post-offices, one-lane toll bridge.

Used formulas:

1. Probability that there are n arrivals in the system $p_n = r^n(1 - r)$

2. Probability of the system being empty: $p_0 = (1 - r)$

3. The expected number in the waiting line and/or being serviced is: $L = \frac{\lambda}{\mu - \lambda}$

4. The expected number in the queue is: $L_q = \frac{\lambda^2}{\mu(\mu - \lambda)}$

5. The average waiting time (in the queue) of an arrival is: $W_q = \frac{L_q}{\lambda} = \frac{L}{\mu}$

6. The average time an arrival spends in the system (both waiting and in service) is: $W = W_q + 1/\mu$

Solution:

The average utilisation of the teller is $\rho = 75$ percent

The average number in waiting line is $L_q = 2,25$ customers

The average number of customers in the system is 3 customers

Average waiting time in waiting line is 0,15 hour e.g. 9 minutes

Average waiting time in the system is 0,2 hour e.g. 12 minutes.

A Multiple-Server Queuing Model

The multiple-server, or multiple-channel model, as it is sometimes referred to, is very similar to the single-server model, except that the number of servers is not limited to one.

Basic properties of the multiple-server model

Suppose that these conditions are satisfied:

1. A Poisson arrival rate.
2. An exponential service time.
3. There are S servers in the system.
4. The processing order is FCFS (first-come, first-served).
This can be described by Kendal classification as system M/M/S/FCFS.
5. The calling population is infinite.
6. There is no upper limit on queue length.
7. All servers are homogeneous - e.g. have the same mean service rate.

Multiple-channel systems are found in banks and post offices where there are single waiting lines and at bakery and deli counters where customers take a number and then are served in that order. Usually, supermarket checkouts would not fall into this category. even though they have multiple checkouts, because customers in a general way do not form a single line.